

PETROSANI, Romania

WMO: 152960

Lat: 45.4064N

Lon: 23.3767E

Elev: 609

StdP: 94.22

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.7	-15.1	1.1	-12.6	-12.5	1.4	-9.7	7.6	2.6	6.1	0.3	0.6	140	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	29.1	20.2	27.6	19.7	26.2	19.2	21.8	27.2	20.7	25.9	19.8	24.8	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.9	24.4	19.0	14.8	23.5	18.1	14.0	22.2	66.8	27.2	62.7	25.8	59.6	24.8	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.4	6.0	4.4														
(5)			DB	-17.6	31.5	2.7	2.0	-19.5	32.9	-21.1	34.1	-22.7	35.3	-24.6	36.7	
			WB	-17.1	24.2	2.1	1.7	-18.7	25.4	-19.9	26.4	-21.1	27.4	-22.6	28.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-1.5	1.2	3.9	9.0	13.5	17.6	18.5	18.7	14.1	9.5	4.5	-0.7	(6)
(7)		DBStd	8.01	3.99	4.08	3.86	3.96	2.99	2.80	2.51	2.50	3.48	3.67	4.19	4.04	(7)
(8)		HDD10.0	1423	356	247	190	63	8	0	0	0	6	54	169	331	(8)
(9)		HDD18.3	3480	614	479	447	281	150	46	29	26	129	274	414	589	(9)
(10)		CDD10.0	1085	0	1	1	32	117	228	264	271	129	38	5	0	(10)
(11)		CDD18.3	100	0	0	0	0	1	23	35	38	3	0	0	0	(11)
(12)		CDH23.3	1221	0	0	0	7	39	273	391	409	96	6	0	0	(12)
(13)		CDH26.7	236	0	0	0	0	2	48	74	102	9	0	0	0	(13)
(14)	Wind	WSAvg	1.1	0.7	1.1	1.3	1.3	1.3	1.2	1.1	1.0	1.0	1.0	0.8	(14)	
(15)	Precipitation	PrecAvg	794	41	38	45	66	89	109	110	67	74	58	52	50	(15)
(16)		PrecMax	1105	83	97	92	122	166	203	221	132	149	139	165	108	(16)
(17)		PrecMin	496	10	6	14	16	30	24	41	14	5	1	0	8	(17)
(18)		PrecStd	157	20	21	24	27	37	46	49	31	37	32	36	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	15.6	19.4	24.3	26.2	30.0	30.1	31.3	28.0	24.3	17.7	11.0	(19)
(20)			MCWB	6.2	9.9	10.8	13.9	18.5	20.6	21.9	20.0	18.3	16.0	12.2	7.4	(20)
(21)		2%	DB	6.9	12.1	16.1	21.6	24.3	28.1	28.7	29.3	26.0	22.0	15.2	8.2	(21)
(22)			MCWB	4.3	8.0	9.1	13.5	17.2	20.3	21.0	19.6	17.9	15.0	11.1	6.0	(22)
(23)		5%	DB	4.8	9.1	13.0	19.4	22.8	26.4	27.3	27.5	24.0	19.1	12.8	6.2	(23)
(24)			MCWB	3.2	6.3	8.1	12.3	16.9	19.6	20.2	19.0	16.6	13.4	10.1	5.0	(24)
(25)		10%	DB	3.3	7.2	10.7	16.8	20.9	24.7	25.9	25.6	21.5	16.4	10.7	4.5	(25)
(26)			MCWB	2.2	5.1	6.7	11.6	16.1	19.0	19.5	18.8	15.8	12.6	8.6	3.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	10.9	11.4	15.9	20.4	22.7	23.7	21.9	20.1	19.0	12.7	8.1	(27)
(28)			MCDWB	9.4	13.2	17.2	20.3	24.7	26.2	27.9	29.2	26.4	22.2	16.9	9.9	(28)
(29)		2%	WB	4.8	8.5	9.9	14.5	18.9	21.5	22.2	20.7	18.3	15.9	11.4	6.3	(29)
(30)			MCDWB	6.3	11.3	14.9	19.8	22.2	26.3	27.4	26.6	24.2	18.5	14.5	7.9	(30)
(31)		5%	WB	3.5	6.7	8.5	13.2	17.5	20.2	20.9	19.8	17.2	14.2	10.3	5.1	(31)
(32)			MCDWB	4.6	8.7	12.6	18.4	21.7	25.0	26.0	25.4	22.4	18.4	12.8	6.1	(32)
(33)		10%	WB	2.3	5.3	7.2	11.9	16.1	19.3	19.9	19.1	16.1	12.6	9.0	3.7	(33)
(34)			MCDWB	3.2	6.9	10.4	16.2	20.2	24.0	24.7	24.5	20.4	16.1	10.6	4.5	(34)
(35)	Mean Daily Temperature Range	MDBR		8.1	9.0	10.1	11.2	11.2	12.1	12.4	12.9	11.5	11.5	8.2	7.2	(35)
(36)		5% DB	MCDBR	10.5	13.3	15.3	17.0	15.5	15.0	15.6	16.3	16.1	17.4	11.6	9.4	(36)
(37)			MCWBR	7.9	9.1	9.2	8.9	9.0	8.1	8.2	7.2	8.0	10.3	7.3	6.9	(37)
(38)		5% WB	MCDWB	9.3	12.6	13.9	15.0	14.7	14.0	14.6	14.2	14.3	15.0	10.8	8.7	(38)
(39)			MCWBR	7.3	8.9	9.0	9.2	9.7	8.5	8.6	7.1	7.9	9.7	7.1	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.305	0.336	0.378	0.440	0.425	0.420	0.437	0.453	0.407	0.368	0.339	0.307	(40)
(41)		taud		2.332	2.247	2.209	2.088	2.199	2.252	2.208	2.158	2.257	2.364	2.386	2.372	(41)
(42)		Ebn at Noon		794	831	843	818	846	850	829	798	804	789	748	752	(42)
(43)		Edh at Noon		78	103	123	152	140	134	138	140	117	91	75	68	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.33	2.12	3.23	4.36	5.19	5.84	5.77	5.23	3.73	2.60	1.53	1.09	(44)
(45)		RadStd		0.17	0.27	0.45	0.48	0.47	0.56	0.48	0.51	0.38	0.30	0.19	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)		+0.64	N/A	N/A	+0.89	N/A	N/A	-114	-168	+91	+44

Nomenclature: See separate page